Design and Simulation of a Photonic Crystal Based Optical D Flip-Flop Using OptiFDTD

Mid-Sem Project(EC839) Report

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Abstract

Photonic crystals are artificial periodic dielectric structures that can control the propagation of electromagnetic waves by creating a photonic band gap (PBG), within which light of certain frequencies cannot propagate. This property makes them highly suitable for compact optical logic devices and integrated photonic circuits. In this project, a two-dimensional photonic crystal based optical D flip-flop is designed and studied using the OptiFDTD simulation platform.

The proposed structure uses a hexagonal lattice of silicon rods in air. By removing selected rods, defect waveguides are created to guide the incoming optical signals. Two input ports corresponding to the data signal (D) and clock signal (CLK), and two output ports corresponding to Q and Q' are considered in the design. The operation of the device is based on optical interference and controlled light propagation through the defect paths and scatter rods placed inside the structure.

The work focuses on understanding the design methodology, photonic band gap analysis, and implementation of the structure in OptiFDTD. The lattice parameters are chosen based on the operating wavelength in the C-band near 1.55 µm. The structure is first analyzed through the PBG concept to identify the suitable wavelength region for operation, and then modeled as a D flip-flop using defect engineering.

At the current stage, the complete optical signal propagation behavior for all input combinations has not yet been fully verified through final simulation plots. However, the structure creation, parameter selection, source placement, detector placement, and simulation setup have been completed successfully. The project forms a strong basis for future completion of field propagation plots, output truth table validation, and performance comparison with reported research results.

CERTIFICATE

I hereby certify that the Project entitled Design & Simulation of Photonic Crystal based optical D-Flip flop using OPTI FDTD, which is being submitted to the **National Institute of Technology Karnataka, Surathkal,** in partial fulfilment of the requirements for the award of the Degree of **B.Tech ECE** in **Department of Electronics and Communication Engineering** is a bonafide report of work carried out by me.

Name: Sanket Sudhakar Sakhare

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Signature

Place: NITK, Surathkal

Date: 08-03-2026

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Name: R. Krithik Kumar

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1. Introduction

Modern communication and computing systems increasingly demand higher speed, larger bandwidth, and lower power consumption than what conventional electronic circuits can offer. As the performance of electronic devices approaches physical scaling limits, optical and photonic technologies have emerged as strong alternatives for future high-speed information processing systems. Optical devices offer significant advantages such as low transmission loss, immunity to electromagnetic interference, high parallelism, and extremely fast data transfer.

Photonic crystals are periodic dielectric structures that can manipulate the flow of light in a controlled manner. Due to their periodic refractive index profile, they exhibit a photonic band gap, which is a frequency region in which electromagnetic waves cannot propagate through the structure. By introducing defects into an otherwise periodic crystal, localized and guided optical modes can be created. This makes photonic crystals useful for realizing optical waveguides, resonators, filters, switches, multiplexers, and logic circuits.

Sequential optical logic devices are especially important for optical computing because they provide memory functionality in addition to pure combinational operation. Among them, the D flip-flop is one of the most fundamental building blocks. In digital systems, a D flip-flop stores the input data at a clock transition and produces a stable output until the next valid triggering event. Implementing such a device in the optical domain is valuable for future all-optical data storage, signal regeneration, buffering, and optical computing architectures.

Recent research has shown that photonic crystal based flip-flops can be realized using interference effects, waveguide coupling, resonant cavities, and nonlinear or linear optical mechanisms. In this project, a two-dimensional hexagonal photonic crystal structure composed of silicon rods in air is selected. The design is inspired by reported research on all-optical D flip-flops operating around 1.55 µm. The optical characteristics are studied using the finite-difference time-domain (FDTD) method through the OptiFDTD software package.

The present work primarily focuses on the design methodology, parameter selection, construction of the photonic crystal lattice, introduction of defects, placement of sources and detectors, and preliminary analysis of the proposed structure. The PBG concept is

used to justify the chosen operating wavelength. Although the final signal propagation plots for the complete D flip-flop operation are still under progress, the device modeling process has been completed and provides a clear path for final implementation and validation.

1.1 Objectives

The objectives of this project are listed below:

- To study the basic theory of photonic crystals and the concept of the photonic band gap.
- To understand the working principle of a D flip-flop in the optical domain.
- To design a two-dimensional photonic crystal lattice using silicon rods in air.
- To identify suitable structural parameters such as lattice constant, rod radius, and operating wavelength based on reported literature.
- To create defect waveguides by removing rods from the periodic structure.
- To place input planes and observation planes appropriately in OptiFDTD.
- To set up the FDTD simulation environment for electromagnetic analysis.
- To obtain the photonic band gap characteristics of the periodic structure.

1.2 Background of D Flip-Flop

A D flip-flop is a memory element that stores one bit of information. It has one data input, one clock input, and typically two outputs, namely Q and Q'. The output Q follows the data input only when the clock signal is active, whereas for inactive clock conditions the output retains its previous state. Thus, the D flip-flop is widely used in registers, counters, state machines, and memory circuits.

In an optical implementation, the same logical function must be achieved using light instead of electrical currents and voltages. The challenge lies in achieving controlled routing, confinement, and interaction of optical signals inside a compact structure. Photonic crystals are attractive for this purpose because defect engineering allows very precise control of optical paths, while the PBG prevents unwanted leakage of light into the surrounding periodic medium.

1.3 Photonic Band Gap Concept

The photonic band gap is one of the central concepts in photonic crystal analysis. It refers to a range of normalized frequencies or wavelengths for which electromagnetic waves cannot propagate through a perfectly periodic photonic crystal. If a defect is introduced into the structure, light can then be guided or localized within that otherwise forbidden region. This property is critical in designing photonic crystal logic devices, because it allows strong confinement of optical signals and reduces power leakage.

In the selected design, the operating wavelength is around 1.55 µm, which lies in the optical communication C-band. The photonic band gap study is therefore important to confirm that the chosen lattice geometry can support the intended device operation near this wavelength.

2. Photonic Crystal D Flip-Flop Structure

The proposed structure is a two-dimensional photonic crystal formed by a hexagonal array of silicon rods in air. Silicon is chosen because of its high refractive index contrast with air, which helps in obtaining a strong photonic band gap and better optical confinement. The structure contains defect regions created by removing selected rods to form waveguide paths for the optical signals.

The design includes two input ports corresponding to the data input (D) and clock input (CLK), and two output ports corresponding to Q and Q'. In addition, reduced-radius scatter rods are introduced inside the defect region to manipulate the optical path and improve switching behavior through interference effects. The overall operation is intended to direct the incoming optical energy into the appropriate output path depending on the combination of the applied D and CLK signals.

The implementation was carried out in OptiFDTD. The software uses the finite-difference time-domain method to solve Maxwell's equations in both time and space domains, allowing observation of electromagnetic field propagation inside the photonic crystal device.

2.1 Design Parameters

The design parameters selected from the reference paper are summarized in Table 2.1.

Table 2.1: Design parameters of the proposed photonic crystal D flip-flop

Parameter	**Value**
Lattice constant	0.8 µm
Dielectric material refractive index	3.47
Operating wavelength	1.55 µm
Lattice type	Hexagonal
Lattice dimensions	20×14
Radius of rods	0.2 µm
Radius of scatter rods	0.1 µm
Input power	1 µW

2.2 Structure Description

The full photonic crystal is first constructed as a perfect periodic lattice. Then selected rods are removed to create two defect waveguide paths for the D and CLK inputs, along with corresponding output paths. Scatter rods are inserted near the interaction region to alter the propagation path using interference and scattering effects. The resulting geometry is intended to function as an optical D flip-flop.

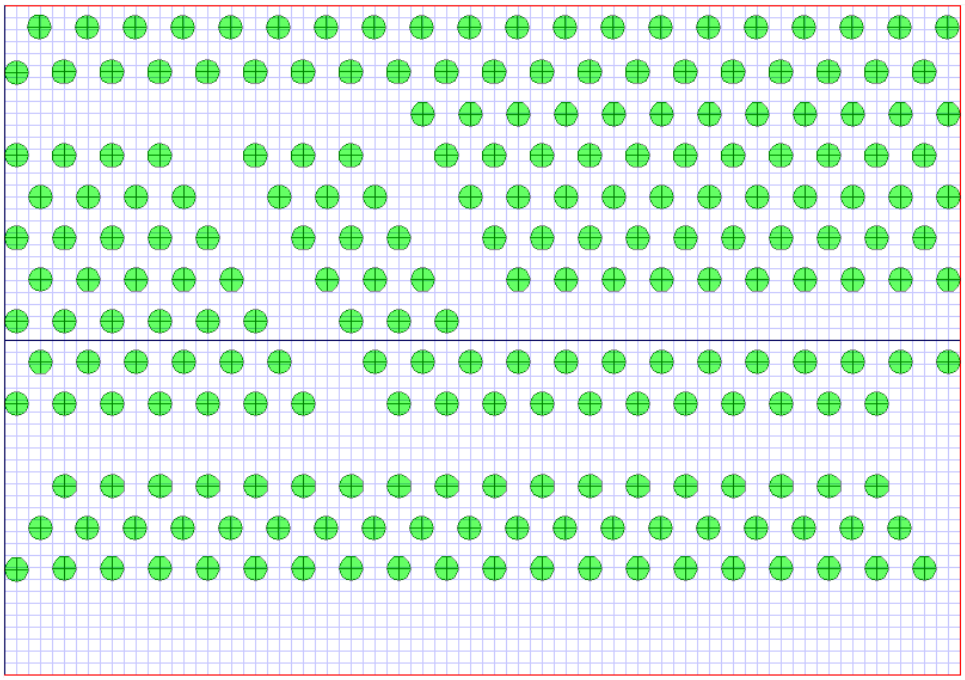

Figure 2.1: Proposed photonic crystal based D flip-flop structure.

2.3 Input and Output Arrangement

Two input planes are used in the simulation setup:

- the D input plane,
- the CLK input plane.

Similarly, two observation planes are placed at the right side of the structure to capture the optical response corresponding to:

- output Q,
- output Q'.

The input planes are placed in the free-space defect channels, not on the rods, so that the incident optical field can efficiently couple into the waveguide path. The observation planes are placed at the output defect channels to record the transmitted optical field and output power.

3. Simulation Methodology

3.1 Software Tool

OptiFDTD was used to design and simulate the proposed structure. It is based on the FDTD method, which discretizes both space and time to solve Maxwell's curl equations numerically. This makes it suitable for studying wave propagation in complex dielectric structures such as photonic crystals.

3.2 Simulation Flow

The main steps followed in the simulation are as follows:

1. Create the two-dimensional simulation domain.
2. Define silicon and air materials.
3. Construct the hexagonal lattice using circular rods.
4. Remove rods to form the defect waveguides.
5. Add scatter rods of smaller radius in the interaction region.
6. Place the D and CLK input planes.
7. Place the Q and Q' observation planes.
8. Set the wavelength, polarization, mesh, and simulation time.
9. Run the simulation and analyze the detector outputs.

4. Results and Discussion

For a photonic crystal based device, the choice of operating wavelength must be supported by the photonic band gap of the periodic structure.

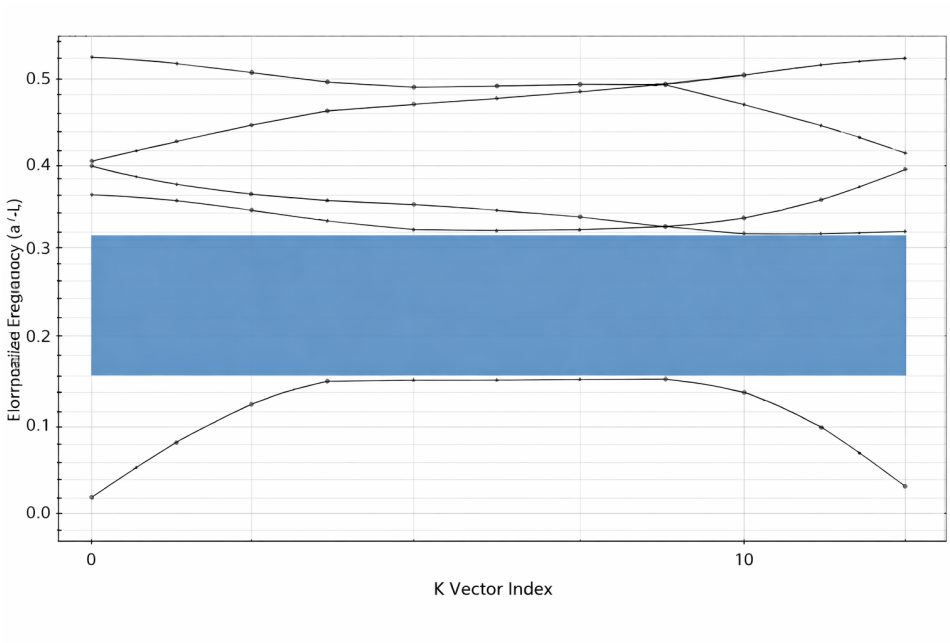

Figure 4.1: Photonic band gap diagram

The band gap region indicates the wavelength range in which propagation through the perfect lattice is forbidden. By introducing defects, guided modes are formed inside that forbidden region, which is the basis for the optical logic operation. The structure creation, defect engineering, and source-detector placement were completed successfully in OptiFDTD. The simulation domain was configured for two-dimensional TE analysis. Input planes and observation planes were placed in the corresponding defect channels. Preliminary simulations confirmed that the software setup and detector arrangement were functioning correctly.

At this stage of the project, the complete set of optical field propagation results for all D flip-flop input conditions is still under progress. The major challenge encountered was accurate source placement, detector sizing, and interpretation of analyzer outputs for a multi-port photonic crystal structure. Even so, the following outcomes have already

been achieved:

- successful creation of the hexagonal photonic crystal lattice,
- formation of the defect waveguide channels,
- inclusion of reduced-radius scatter rods,
- placement of two input planes and two observation planes,
- preliminary detector response generation in OptiFDTD Analyzer,
- understanding of the PBG role in wavelength selection.

4.1 Expected Signal Propagation Behavior

Based on the reference design, when the clock signal is active, the structure should direct the data signal toward the appropriate output path. The outputs Q and Q' are expected to exhibit complementary behavior depending on the applied D input. The functionality is based on controlled interference and confinement of light inside the defect channels.

The final validated propagation images for the main cases such as CLK = 1, D = 0 and CLK = 1, D = 1 are planned as part of the completion stage of the work. These results will allow direct comparison between the implemented OptiFDTD structure and the reported research paper.

5. Conclusion and Future Work

This project focused on the design and preliminary simulation of a photonic crystal based optical D flip-flop using OptiFDTD. A two-dimensional hexagonal lattice of silicon rods in air was constructed, and defect waveguides were introduced by removing selected rods. Scatter rods were added to control signal propagation through interference-based mechanisms. The structure was designed with two input ports, namely D and CLK, and two output ports, namely Q and Q'.

The main contribution of the work completed so far is the successful translation of a reported photonic crystal D flip-flop design into a practical OptiFDTD model. The structural parameters, source placement methodology, detector placement, and simulation settings were studied and implemented. The theoretical basis of operation through the photonic band gap and defect-guided propagation was also established.

Although the complete optical signal propagation plots and final output verification are still pending, the developed model forms a complete foundation for the remaining work. The next stage of the project will focus on obtaining final field distribution plots for different combinations of D and CLK inputs, validating the truth table, and comparing the observed performance with the values reported in the research paper.

Future work will include:

- completing the optical signal propagation analysis for all required D flip-flop states,
- extracting clearer detector response plots from OptiFDTD Analyzer,
- refining the detector and source placement for improved accuracy,

Thus, the project has reached a strong intermediate stage and can be completed into a full simulation study with additional propagation analysis and detector-based verification.

Bibliography

[1] S. Damodaran, T. Shankar, and R. Anbazhagan, "All optical clocked D flip flop for 1.72 Tb/s optical computing," *Microelectronics Journal*, vol. 103, p. 104865, 2020.

[2] T. S. Mostafa and E. S. M. El-Rabaie, "All-Optical D-Flip Flop with Multi-Wavelength Operation Based on Photonic Crystal," *International Japan-Africa Conference on Electronics, Communications, and Computations*, 2019.

[3] F. Paradin, A. Sheykhian, and A. Askarian, "A novel design of an all-optical D flip-flop based on 2D photonic crystals," *Journal Name*, vol. 66, no. 1, p. e34006, Jan. 2024.

[4] K. Sakoda, *Optical Properties of Photonic Crystals*. Springer, 2005.

[5] J. D. Joannopoulos, R. D. Meade, and J. N. Winn, *Photonic Crystals: Molding the Flow of Light*. Princeton University Press, 1995.

[6] E. Yablonovitch, "Inhibited spontaneous emission in solid-state physics and electronics," *Physical Review Letters*, vol. 58, no. 20, pp. 2059–2062, 1987.

[7] Optiwave Systems Inc., "OptiFDTD User Reference Guide." Available online: `https://optiwave.com`